第六周

簡述學習目標

- (open web ui)“看” 資料：了解資料品質如何影響模型的表現，學習從 hugging face 等平台辨識高品質資料集，並能根據需求自行設計資料
- (hugging face)“用” 資料：學習將資料轉成外部知識庫，透過 RAG 讓模型依據真實資料進行回覆
- “擴充”&”校準”資料：了解合成資料在資安稀缺樣本下的角色，學習如何根據原來的資料集生出更多資料，並審核 AI 產出的邏輯，建立高品質的輸出

TASK 1 (Achieving)：資料集蒐集與整理

作業內容

1. 主題資料收集
    - 在 hugging face 等平台搜尋相同主題資料集，觀察其資料架構與格式：專業資料集多用了哪些你原本沒想到的欄位？ 這些欄位對模型判斷可能有什麼幫助？
    - 選擇一個資安相關主題 (可和上禮拜主題一樣)，根據該主題設計你的資料集的架構，並說明 schema 如何幫助模型理解資料
    - 根據設計的欄位，收集、清洗並整理出一份初步的資料集
2. 知識庫應用
    - 使用 open web ui，將你的資料集結構化資料轉成 markdown 或帶標籤的段落放入知識庫
    - 比較掛載知識庫前與掛載知識庫後針對同一個問題的回覆差異，觀察模型是否能根據真實資料精準回答

截圖內容

- 欄位設計說明
- 對比的資料集
- Open web ui 知識庫設定
- 掛載知識庫前後對比

TASK 2 (Exceeding)：RAG 流程程式實作

作業內容

1. (hugging face)延續 task1 主題，寫程式實作 RAG 的運作流程，並了解底層邏輯跟步驟(包含資料讀取與切片、向量化、檢索、生成..)

    參考連結：https://huggingface.co/learn/cookbook/zh-CN/advanced_rag

截圖內容

- RAG 實作程式碼截圖
- 程式執行結果截圖

TASK 3 (Outstanding)：synthetic data

作業內容

1. 說明 synthetic data 在資安領域扮演的角色與價值
2. 沿用 task 1 設計的資料集 schema，寫 prompt 引導 LLM 生出 10 - 20 筆合成資料，或者用非 LLM 的演算法方式生出合成資料，說明你這樣設計的理由
3. 品質校準與分析：從生成的資料中選出 1 筆最合理的與 1 筆不合理的 (AI 瞎掰的，或者合成演算法的瑕疵)，並說明你將如何解決這個問題

截圖內容

- 合成資料生成指令或演算法截圖
- 合成資料結果截圖
- 品質校準對比分析截圖

Week 6

Learning Objectives

- **(Open WebUI) "View" Data:** Understand how data quality affects model performance. Learn to identify high-quality datasets from platforms like Hugging Face and design your own datasets based on specific needs.
- **(Hugging Face) "Use" Data:** Learn how to convert datasets into external knowledge bases and use RAG to allow the model to answer based on real data.
- **"Expand" & "Calibrate" Data:** Understand the role of synthetic data in cybersecurity scenarios with limited samples. Learn how to generate additional data based on your existing dataset and review AI-generated outputs to ensure logical consistency and high-quality results.

TASK 1 (Achieving): Dataset Collection and Preparation

Assignment Content

1. Topic Data Collection
    - Search for datasets on the same topic on platforms like Hugging Face and examine their data structure and format. Which fields do professional datasets include that you didn't think of? How might these fields help the model's reasoning?
    - Select a cybersecurity-related topic (it can be the same as last week), design the schema of your dataset based on this topic, and explain how the schema helps the model understand the data.
    - Based on your designed fields, collect, clean, and organize an initial version of your dataset.
2. Knowledge Base Application
    - Using Open WebUI, convert your dataset into structured Markdown or labeled paragraphs and add them to the knowledge base.
    - Compare the model's responses to the same question **before and after connecting the knowledge base**, observing whether the model can answer accurately based on real data.

Screenshots to Include

- Field/schema design explanation
- Compared datasets
- Open WebUI knowledge base settings
- Before-and-after comparison of responses

TASK 2 (Exceeding): RAG Workflow Implementation

Assignment Content

1. Continuing the topic from Task 1 on Hugging Face, implement the RAG workflow through code, understanding the underlying logic and steps (including data reading & chunking, vectorization, retrieval, generation, etc.)

   Reference link: [Hugging Face Advanced RAG Cookbook]

Screenshots to Include

- RAG implementation code
- Program execution results

TASK 3 (Outstanding): synthetic data

Assignment Content

1. Explain the role and value of synthetic data in the cybersecurity field.
2. Using the dataset schema designed in Task 1, write prompts to guide an LLM to generate 10–20 synthetic records, or use a non-LLM algorithm to generate synthetic data. Explain your design rationale.
3. Quality calibration and analysis: From the generated data, select **1 record that is most reasonable** and **1 record that is unreasonable** (AI-generated hallucination or algorithmic flaw), and explain how you would address the issue.

Screenshots to Include

- Synthetic data generation prompts or algorithm screenshots
- Synthetic data results
- Quality calibration and comparison analysis